

BioR5: Biological Reasoning System

A Three-Layer Tool-Calling Architecture

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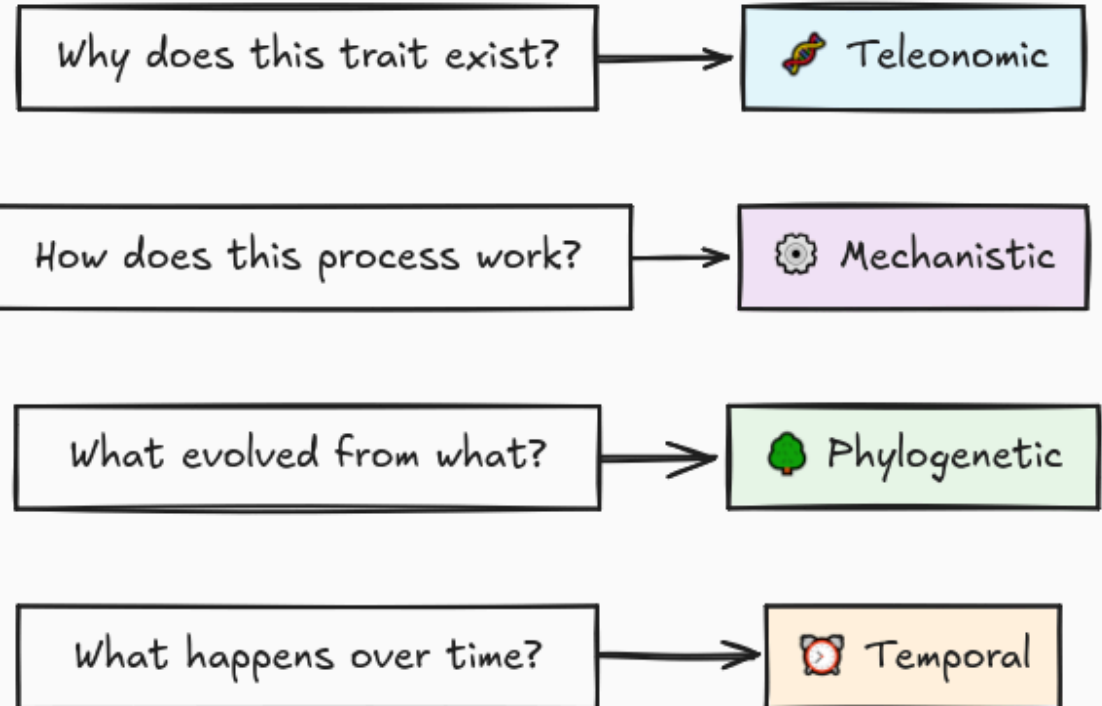
The Problem

The challenge: Biologists use different reasoning modes

- "Why does this trait exist?" → Teleonomic
- "How does this work?" → Mechanistic

Current AI: One model, one approach

Biology: Eleven distinct reasoning modes



Our Solution

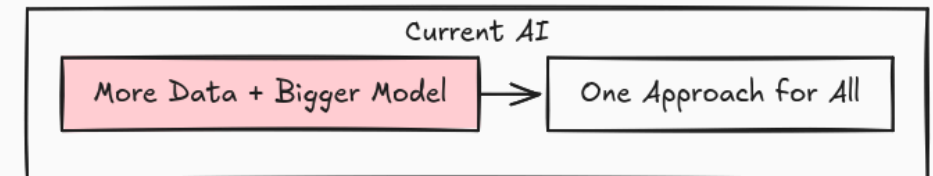
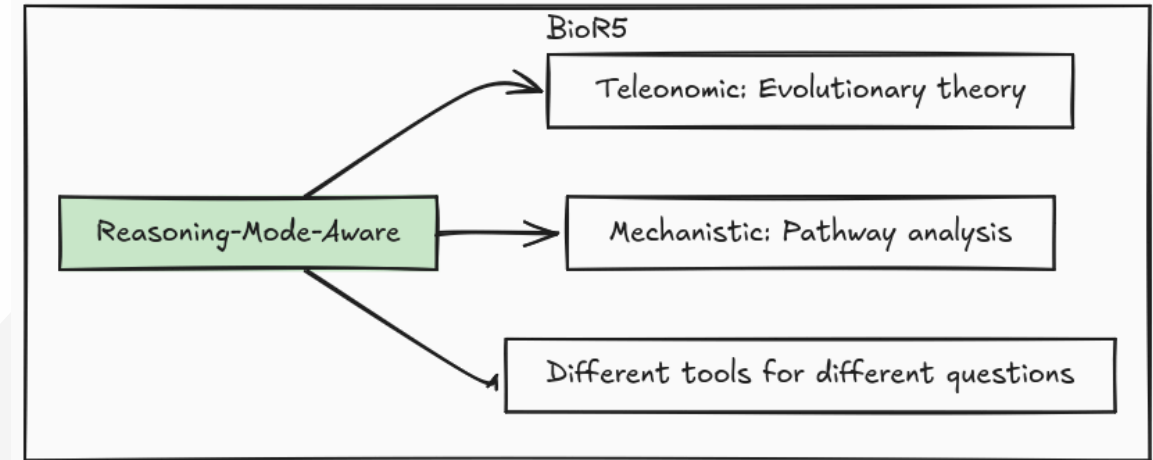
BioR5: Different questions →

Different approaches

Current AI: "More data + bigger model"

Example: "How does insulin regulate glucose?"

- Needs biochemical pathways + causal networks
- NOT just correlation in expression data



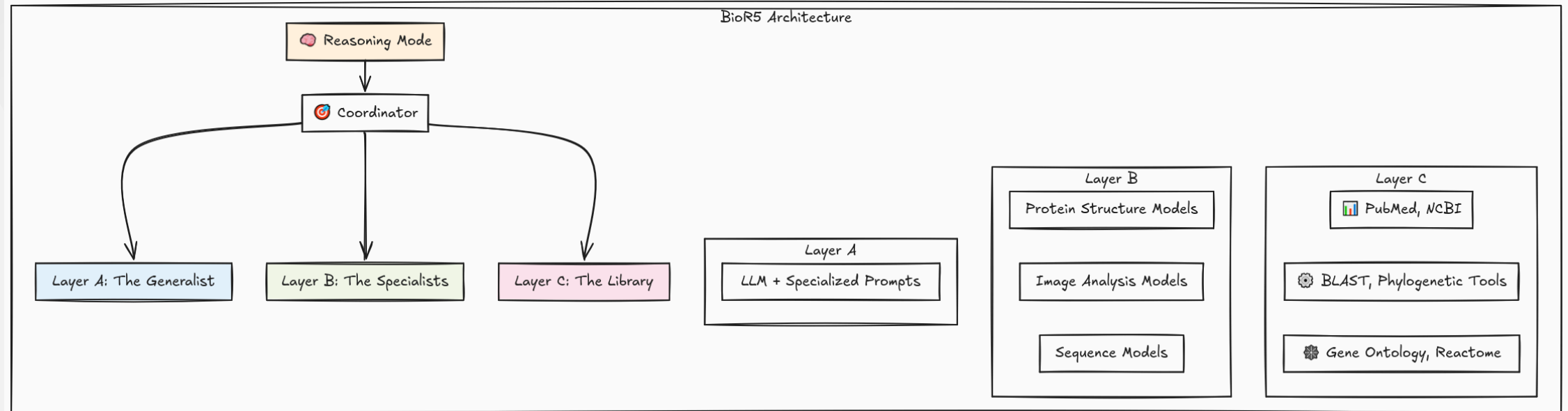
Three-Layer Architecture

BioR5: Map reasoning modes to computational layers

Layer A: LLM + specialized prompts

Layer B: Specialized models (proteins, images)

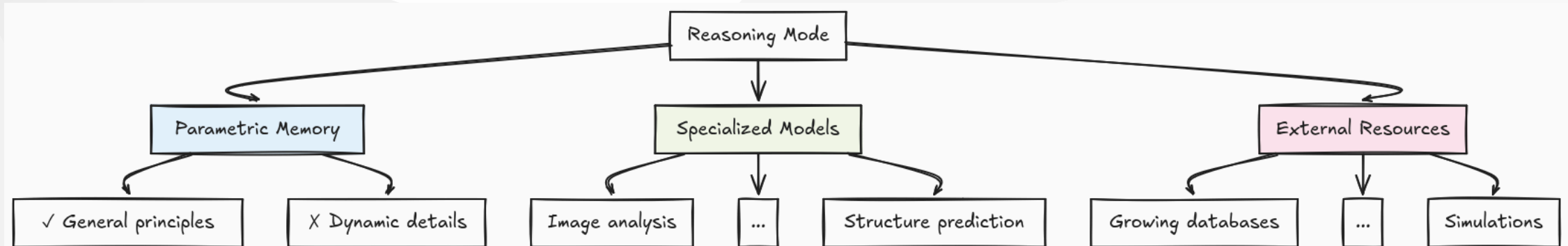
Layer C: External resources (databases, tools)



Reasoning Mode Details - A Recipe

Key Insight: Each reasoning mode needs different resources:

1. **Model weights:** Principles ✓, Details ✗
2. **Specialized models:** Images, structures, etc.
3. **External:** Databases, simulations, etc.



Layer Details - ToolRegistries

Layer A: `parametric_memory` = LLM + specialized prompts

- Same LLM, different prompts → Different knowledge distillation
- Status: 11 reasoning modes implemented

Layer B: Specialized models as tools

- Structure prediction, image analysis
- Packaged as callable tools

Layer C: External resources

- Databases, computational tools, knowledge graphs

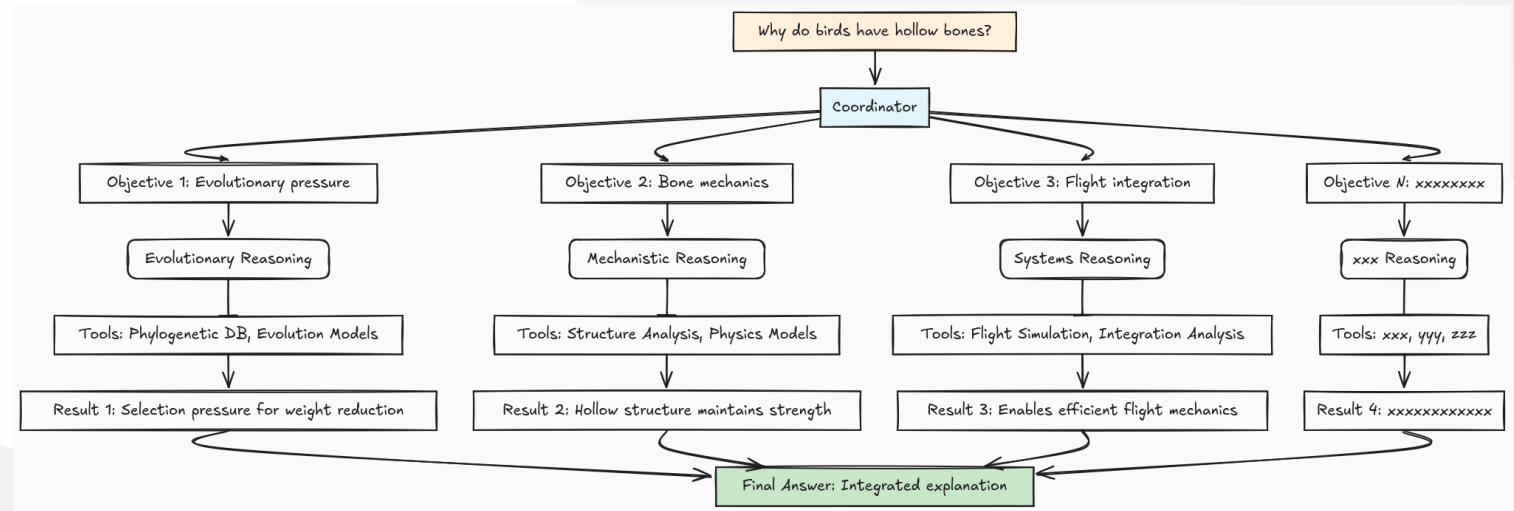
Work in Progress

Next: Divide-and-conquer scheduling

Example: 'bird bone evolution' →

- Evolutionary pressure
- Bone mechanics
- Flight integration

Results merge into final answer



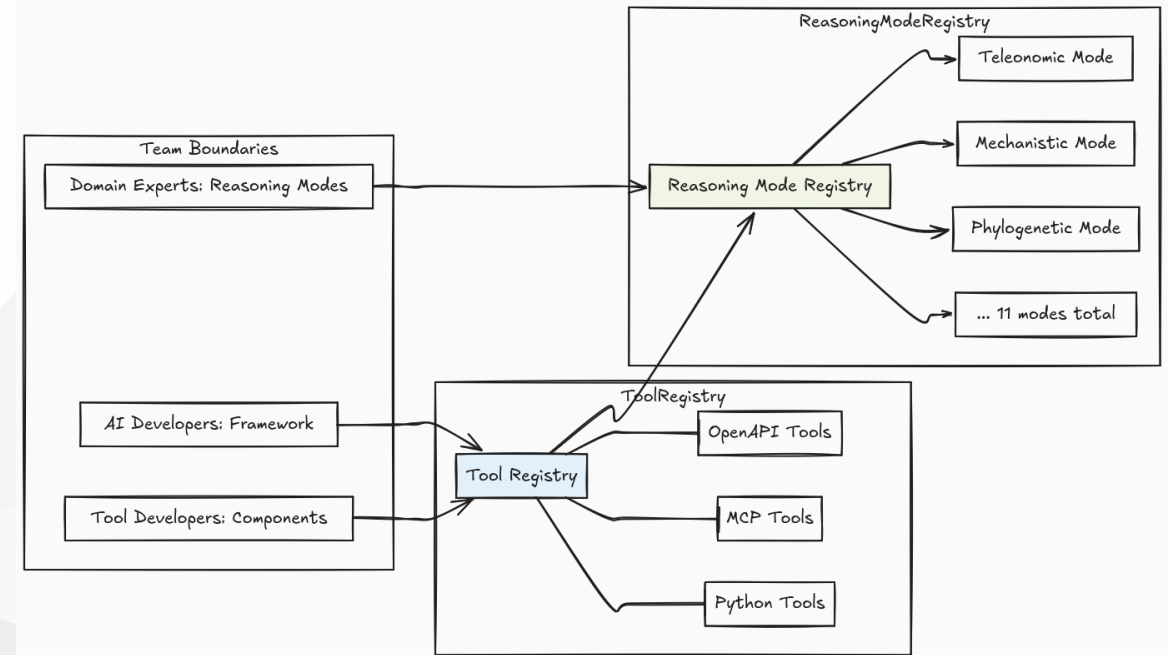
Why This Matters

Scalability:

- ToolRegistry: Any OpenAPI/MCP/Python function becomes a tool
- ReasoningModeRegistry: Recipes for reasoning

Team boundaries:

- AI developers: Framework
- Domain experts: Reasoning modes
- Tool developers: Components



Resources

- BioR5: <https://github.com/brettin/Biological-Reasoning>
- ToolRegistry: <https://toolregistry.readthedocs.io/>

Thank You

Questions?